SIMATS ENGINEERING

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Engineer to Excel

Student Name: S. YASO PRIYA **Reg.No.:192424394**

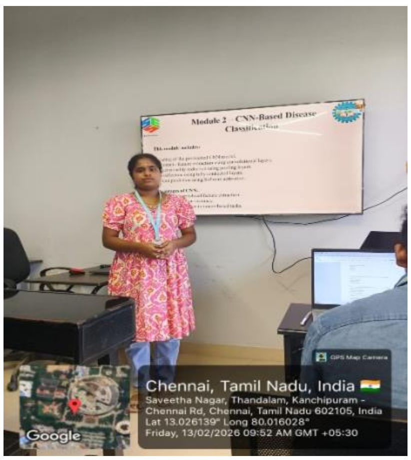
Course Code: DSA0216 **Slot:** B

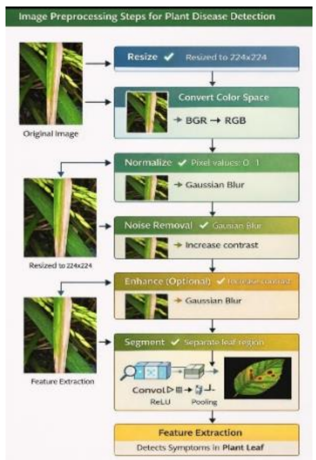
Course Name: **Computer Vision with OpenCV for Modern AI**

Course Faculty: Dr. Senthilvadivu S & Dr. Kumaragurubaran T

Project Title: REALTIME PLANT LEAF DISEASE CLASSIFICATION WITH OPENCV

Module Photographs:





Project Description: Result Interpretation and Reporting

This module focuses on the automatic classification of paddy leaf diseases using a Convolutional Neural Network (CNN). After image acquisition and preprocessing, the processed leaf images are provided as input to the CNN model. The CNN architecture is designed to automatically extract relevant features such as texture patterns, color variations, and lesion structures from the paddy leaf images.

The model consists of multiple convolutional layers, pooling layers, and fully connected layers that work together to learn hierarchical representations of disease characteristics. The extracted features are passed through dense layers, and the final output layer uses the Softmax activation function to classify the image into one of the predefined disease categories.

During training, the model learns from labeled datasets and optimizes its parameters using backpropagation and an appropriate optimization algorithm. After training, the model can accurately classify new, unseen paddy leaf images. The output of this module is the predicted disease label along with a confidence score, indicating the probability of the detected class.

Student Signature **Guide Signature**